

The multiplicity theorem with arbitrary (k, h) , all exponents equal:

$$Z_d(n) \sim \left(\prod_i k_i^{-1} \right) \frac{A_{p-1}}{2^{d-1}(d-1)!} n^{-p/2} (\log n)^{d-1}$$

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Abstract

The d -fold chart standard integral with exponent vectors $k, h : \mathbb{N} \rightarrow \mathbb{N}$ is defined recursively, $Z_0(c) = f(c)$, $Z_{d+1}(c) = \int_0^b u^{h_d} Z_d(cu^{k_d}) du$. When all candidate exponents $p_i = (h_i + 1)/k_i$ coincide at p , it reduces exactly to the iterated weighted primitives of unit 49, $Z_{d+1}(c) = (\prod_{i \leq d} k_i^{-1}) c^{-p} H_d(cb \sum k_i)$ with $H = \text{iterDivPrim}(F_{p-1})$, and hence $Z_{d+2}(\sqrt{n}) \sim (\prod k_i^{-1}) \frac{A_{p-1}}{2^{d+1}(d+1)!} n^{-p/2} (\log n)^{d+1}$: the multiplicity- $(d+2)$ statement of `thm:TaylorTree` for arbitrary (k, h) . Unit 39 is the case $k = h + 1 = 1$. The exact reduction was verified by three-dimensional quadrature at $k = (1, 2, 1)$, $h = (0, 1, 0)$ to $2 \cdot 10^{-16}$. Zero `sorry/axiom`.

1 Setting

Each coordinate contributes one chart power substitution (unit 44) turning $u^h(\cdot)(cu^k)$ into $\frac{1}{k} \int_0^{b^k} s^{p-1}(\cdot)(cs) ds$, and one weighted scaling. At the innermost level the kernel becomes F_{p-1} ; at every further level the factor $(cs)^{-p}$ from the previous level combines with s^{p-1} to s^{-1} , which is exactly the “divide and integrate” step of `iterDivPrim`, with the scale absorbed into $b^{\sum k_i}$.

2 The things formalised

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noncomputable def iterChartGen ( a b : ) (k h : → ) : → →
  | 0 => fun c => quadKernel a c
  | d + 1 => fun c => u in Ioc 0 b, u ^ (h d) * iterChartGen a b k h d (c * u ^ (k d))
theorem iterChartGen_succ_eq ...
  (hp : i, i d → ((h i : ) + 1) / k i = p) (hc : 0 < c) :
  iterChartGen a b k h (d + 1) c
    = ( i Finset.range (d + 1), (1 : ) / k i) * c ^ (-p)
      * iterDivPrim (weightedPrimitive a (p - 1)) d
        (c * b ^ ( i Finset.range (d + 1), k i))
theorem iterDivPrim_isEquivalent (hG : LogBase G A M C ) (hA : 0 < A)
  (m : ) (hm : 0 < m) :
  (fun L => iterDivPrim G m L) ~[atTop] fun L => A / (m.factorial : ) * Real.log L ^ m
theorem iterChartGen_isEquivalent ... (hp : i, i d + 1 → ((h i : ) + 1) / k i = p) :
  (fun n => iterChartGen a b k h (d + 2) (Real.sqrt n)) ~[atTop]
  fun n => ( i Finset.range (d + 2), (1 : ) / k i)
    * (weightedMass a (p - 1) / (2 ^ (d + 1) * ((d + 1).factorial : )))
    * (n ^ (-(p / 2)) * Real.log n ^ (d + 1))

```

3 Proof strategy

The reduction is an induction on d generalising the scale c . The base case is unit 44's inner computation; the step rewrites the inner integral by the induction hypothesis at scale cu^{k_d} , factors $(cu^k)^{-p} = c^{-p}(u^k)^{-p}$, applies the chart power substitution to $u^h g(u^k)$ with $g(s) = s^{-p}H_d(cb^S s)$, simplifies $s^{p-1}s^{-p} = s^{-1}$, and applies the scaling lemma with $\gamma = -1$, whose prefactor $(cb^S)^{-(-1+1)}$ is 1; the result is `iterDivPrim_succ` at the point cb^{S+k_d} . The asymptotic of the iterated primitives of any admissible base follows from unit 49's expansion: the lower-order terms $\sum_{i < m} c_{m,i} \log^i L$ are $o(\log^m L)$ (`isLittle0_pow_pow_atTop_of_lt` composed with `log`, summed with `isLittle0.sum`) and the remainder is $O(1) = o(\log^m L)$. The final assembly is unit 39's template with $(\sqrt{n})^{-p} = n^{-p/2}$.

4 Roadblocks and resolutions

- The scaling lemma at $\gamma = -1$ leaves the exponent $-(-1 + 1)$, not $-1 + 1$; the `show` rewrite must match the negated form before `Real.rpow_zero`.
- `isLittle0_pow_pow_atTop_of_lt` lives in `SpecificAsymptotics`, which was not in the import closure; import it explicitly.
- `isLittle0.sum` produces a `Finset`-sum of functions; convert with `Finset.sum_apply` in a `congr_left`.

5 What was Mathlib, what was new

Mathlib: `isLittle0_pow_pow_atTop_of_lt`, `isLittle0.sum`, `tendsto_pow_atTop`, `Finset.prod_range_succ`, `Finset.sum_range_succ`, `Real.mul_rpow`. New: the `general-(k,h)` chart recursion, its exact reduction, and the multiplicity theorem in the all-equal case with arbitrary exponents.

6 Lessons

The abstraction of unit 49 paid off immediately: nothing about the base F_{p-1} had to be reproved, and the `general-(k,h)` recursion is the same three-line pointwise identity at every level. Verifying the exact reduction numerically in three dimensions before writing the induction gave confidence in the bookkeeping of $\prod k_i^{-1}$ and $b^{\sum k_i}$.

7 Outcome

Units 44–50 lift the whole leading-term theory of `thm:TaylorTree` from $k = h + 1 = 1$ to arbitrary (k,h) : two dimensions in all three regimes, and d dimensions when all exponents coincide. What remains at this level of the paper is the mixed case in general dimension (some coordinates at the minimal exponent, others above), whose two-dimensional and $d = 3$ instances (units 37, 40, 46) suggest the answer: the minimal coordinates alone determine the logarithmic power, the others contribute bounded factors.